

MEDICINE

Solving the puzzle of Long Covid

Long Covid provides an opportunity to understand how acute infections cause chronic disease

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More than 4 years into the global COVID-19 pandemic, widespread infection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) has left millions of people around the world with Long Covid, which describes the constellation of post-acute and long-term adverse health effects caused by the infection. Evidence generated by the scientific community—with formidable contributions from patient-led research teams—has provided a thorough understanding of the epidemiology and clinical manifestations of Long Covid. Understanding the biologic underpinnings of this disease is also improving, along with evidence that vaccination and antivirals can help prevent it. Yet despite this progress, prevention efforts have stalled, there is uncertainty about governments' long-term commitment to address research needs in this area, and there has yet to be a treatment option validated with randomized controlled trials (RCTs).

Long Covid can manifest in people across the life span (from children to older adults) and across race and ethnicity, sex, and baseline health status. It is a complex nonmonolithic multisystemic disease with sequelae across almost all organ systems. Long Covid is likely a disease with many subtypes that may have different risk factors (genetic, environmental, etc.) and distinct biologic mechanisms that may respond differently to treatments. For example, the prototypical (classic) form of Long Covid (with brain fog, fatigue, dysautonomia, and postexertional malaise) is more common in younger adults and in females. Other forms of Long Covid, including those with cardiovascular and metabolic sequelae, are manifest more often in older adults and those with comorbidities. A common risk across all types of Long Covid is severity of acute infection; the risk—on the relative scale—increases according to the severity of the acute infection. However, despite the lower relative risk, more than 90% of cases occur in people who had

mild SARS-CoV-2 infection, owing to the much higher prevalence of mild cases (1).

From an extensive body of mechanistic research in people affected by Long Covid, there appear to be multiple potential pathogenic pathways, including persistence of the virus or its components in tissue reservoirs; autoimmune or an unchecked, dysregulated immune response; mitochondrial dysfunction; vascular (endothelial) and/or neuronal inflammation; and microbiome dysbiosis (2). In people with severe COVID-19, systemic acute infection can arise in which SARS-CoV-2 replicates in pulmonary and extrapulmonary tissue, and its genomic RNA may persist for months in multiple sites, including the brain and coronary arteries (3). To what extent this happens in milder cases and whether this contributes mechanistically to Long Covid is not yet clear. SARS-CoV-2 may also reactivate dormant viruses, including Epstein-Barr virus and varicella zoster virus, as well as lead to gut-brain and neuroendocrine dysfunction, mitochondrial dysfunction, and impaired coagulation (4, 5) (see the figure).

Because nonpharmaceutical interventions to reduce the risk of SARS-CoV-2 transmission have largely been abandoned, vaccines are now the primary line of defense against both severe disease in the acute phase of the infection and Long Covid. Studies have consistently shown that vaccines reduce the risk of Long Covid by 15 to 75%, with a mean of ~40% reduction in risk. Yet vaccine policies in much of the world restrict boosters to older adults or those with risk factors for severe COVID-19, and with pandemic fatigue, the public's appetite for boosters seems to be waning.

Evidence from observational analyses also suggests that use of the antiviral ritonavir-boosted nirmatrelvir within 5 days of symptom onset of SARS-CoV-2 infection may reduce the risk of Long Covid by 26% (6). Molnupiravir—with known mutagenic effects on SARS-CoV-2 and putative mutagenic potential in human cells—has also been shown to reduce the risk of Long Covid. However, the studies on ritonavir-nirmatrelvir and molnupiravir were limited to those who presently qualify for antivirals, including older adults or those with risk factors for the progression to severe COVID-19, and the mutagenic potential of molnupiravir is concerning because

mutations of SARS-CoV-2 may spawn new variants and mutations in humans may lead to cancer or other diseases. Another SARS-CoV-2 antiviral, ensitrelvir, has been shown to reduce the risk of Long Covid in preliminary analyses. Additionally, the diabetes drug metformin initiated within 7 days of SARS-CoV-2 infection reduced the risk of Long Covid by 41% in a RCT (7). More evidence is needed to evaluate the effectiveness of reducing Long Covid risk and the safety of various antivirals, especially in younger and healthier adults (where evidence is lacking), and to reproduce the findings on metformin. Whether a combination of metformin and an antiviral, a combination of antivirals, or a higher dose or longer duration of treatment with an antiviral would yield greater effectiveness in reducing the risk of Long Covid should also be investigated.

Reinfection, which is now the dominant type of SARS-CoV-2 infection, is not inconsequential; it can trigger *de novo* Long Covid or exacerbate its severity. Each reinfection contributes additional risk of Long Covid: Cumulatively, two infections yield a higher risk of Long Covid than one infection, and three infections yield a higher risk than two infections (8). Whether different SARS-CoV-2 variants alter the risk of developing Long Covid should be investigated. Regardless, efforts to prevent reinfection are important and may reduce the risk of long-term population health loss (8).

Despite this cumulative knowledge on mechanisms, epidemiology, and prevention, there are several major challenges. Importantly, the care needs of people with Long Covid are unmet. Patients are often met with skepticism and dismissal of their symptoms as psychosomatic. The attribution of symptoms to psychological causes has no scientific support; it perpetuates stigma and disenfranchises patients from accessing the care they need. Multidisciplinary Long Covid clinics, if they exist at all, are stretched beyond capacity, and they are nearly nonexistent in low- and middle-income countries. The care needs of people with Long Covid around the world must be addressed. This requires training health care providers to recognize and manage Long Covid, expanding access to specialty clinics, and developing care pathways that could be adapted in low-resource settings.

The lack of consensus on terms, definitions, and clinical trial end points for Long Covid is slowing progress and hampering industry engagement in clinical trials. Building consensus on these parameters—for applications that include clinical care, epidemiology and surveillance, clinical trials, and health services research—is urgently needed.

Evidence from multiple studies with 2 to 3 years of follow-up indicates prolonged risk

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for many sequelae (9) and that spontaneous recovery or return to baseline status is uncommon. Extended follow-up of the 1918 influenza pandemic, poliomyelitis outbreaks, and Epstein-Barr virus infections has demonstrated that new, disabling sequelae of these infections can occur multiple decades later; it is uncertain whether this will also occur with COVID-19 (10). This highlights the need for studies to characterize the health trajectories of Long Covid over a longer timeline and allow the identification of latent effects that may not have materialized.

Investment in developing animal models that recapitulate the breadth of Long Covid phenotypes is critical to deepen mechanistic understanding of how SARS-CoV-2 infection leads to chronic disease(s) and to screen po-

addressing the care needs of people suffering from this condition. COVID-19 vaccines are safe and effective, but like any pharmaceutical agent, they are not free of adverse events. Assessment of prevalence, identification of at-risk groups, and research into the underlying mechanisms of vaccine adverse events should be undertaken. Because of the putative overlap in some of the clinical manifestations, understanding the mechanisms of vaccine injury may offer insights into mechanisms of Long Covid and identify potential druggable targets.

The pandemic has laid bare a blind spot in epidemiology and surveillance data systems for infectious diseases. Nearly all surveillance data systems are built on the archaic, and now obsolete, notion that accounting

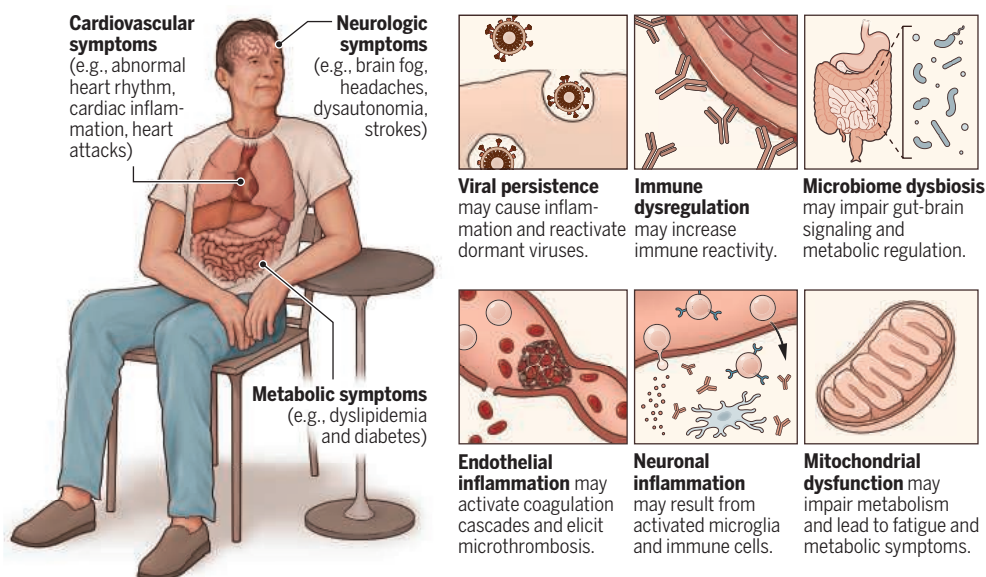
in addition to the prototypical form of Long Covid, SARS-CoV-2 infection increases the risk of a wide array of chronic diseases and will contribute to a rise in the burden of cardiovascular disease, diabetes, neurologic impairment, and autoimmune conditions. This will increase demand on health systems and raise health care costs, and their impact may take many years to fully manifest. Long Covid affects the development and educational attainment of children and reduces labor participation and economic productivity in working-age adults. Both the direct effect of increased risk of death in people with Long Covid and the indirect effect on mortality through increased burden of chronic diseases caused by SARS-CoV-2 (e.g., cardiovascular, metabolic, and neurologic diseases) may contribute to further decline in life expectancy, potentially erasing decades of progress. Addressing these challenges will require continued focus on strengthening prevention efforts and urgently finding treatments for Long Covid.

Finding treatments for Long Covid must be prioritized. The present response to this disease must match the urgency and scale of the problem. Registered trials for Long Covid are too small and underpowered to yield conclusive evidence; they are being conducted at a slow pace that is unlikely to yield answers to inform patient care for years, if not decades. There is no shortage of attractive candidate interventions, many of which are repurposed drugs, that need to be properly assessed. New treatments should also be developed, and approaches—including microbiome modulation, which has shown promising results in one RCT (NCT04950803)—should be further evaluated. Well-powered and rigorously designed RCTs with innovation, such as incorporating digital, site-less, remote enrollment and assessment, are urgently needed to enable rapid enrollment at scale to test the efficacy and safety of a long list of candidate drugs and nonpharmaceutical interventions (2, 11).

Preventing infections and reinfections is the best way to prevent Long Covid and should remain the foundation of public health policy. A greater commitment to nonpharmaceutical interventions, which include masking, especially in high-risk settings, and improved air quality through filtration and ventilation, are requisite. Updating building codes to require mitigation against airborne pathogens and ensure safer indoor air should be treated with the same seriousness afforded to mitigation of risks from earthquakes and other natural hazards. Reducing the risk of serious outcomes after COVID-19 and some prevention of Long Covid can be attained with vaccination of a wider spectrum of the population. Given the dwindling

Many forms of Long Covid

Long Covid is a multisystemic disease with sequelae that affect almost all organ systems. Various putative mechanisms that underlie these sequelae are not mutually exclusive and may explain the myriad health effects seen in Long Covid. Therapeutics that target these pathways, such as antivirals, anti-inflammatory agents, microbiome restoration, and anticoagulant drugs, may ameliorate symptoms.



tential therapeutics. Because Long Covid is a nonmonolithic heterogeneous entity, it is unlikely that a single or a few biomarkers will explain its vast complexity. A creative rethinking of the approach for biomarker development by harnessing the power of artificial intelligence to analyze large numbers of variables and identify multidimensional biomarkers that can classify distinct Long Covid subtypes and predict prognosis and treatment responses is needed.

Tied with the antiscience, antivaccine movement, a tide of Long Covid denialism is rising. This movement sows doubt about the scale and urgency of Long Covid, conflates Long Covid with vaccine adverse events, and seeks to hamper progress on

for cases, hospitalization, and death in the acute phase is sufficient to capture the health burden of the infection. This approach does not account for the burden of long-term health loss due to infectious illnesses, which obscures their true toll. Adding to this challenge are the absent, underdeveloped, or siloed health care data systems in much of the world. This calls for the need to redesign surveillance data systems for infectious agents and to devise solutions for health care data integration to enable the expeditious generation of high-quality evidence in near real time to address pressing knowledge gaps in the setting of public health emergencies.

Long Covid will have wide-reaching effects that are yet to be fully appreciated. In

appetite for COVID-19 boosters, strategies to improve uptake (e.g., pairing it with the annual influenza vaccine) may be effective. Development of more durable, variant-proof vaccines that are not vulnerable to evasion by the ever-mutating virus needs to be accelerated (12). Nasally or orally administered vaccines that induce strong mucosal immunity to block infection and transmission should be pursued, and there are preliminary supportive data from clinical trials (10). It is also necessary to broaden the pipeline of SARS-CoV-2 antivirals, especially because of rising resistance (13).

Governments and funding agencies should support a comprehensive portfolio of research in infection-associated chronic illnesses. Although SARS-CoV-2 can cause chronic disease, it is not alone. Influenza virus, Epstein-Barr virus, Ebola virus, polio virus, and many others also have long-term health effects (14, 15). Myalgic encephalomyelitis/chronic fatigue syndrome is also likely triggered by infection. However, how infectious agents cause chronic disease is not fully understood, and research in this area has been disproportionately underfunded relative to the burden of long-term disability and disease caused by infections. An interdisciplinary research agenda is essential to address knowledge gaps about such illnesses. This effort will not only provide insights into infection-associated chronic illnesses similar to Long Covid but also inform strategies to reduce the burden of chronic health loss from future pandemics.

The world has already paid an extraordinarily heavy price for the biggest pandemic of the 21st century so far. In this crisis lies an urgent need to address the challenge of Long Covid. There is also a historic opportunity—that must not be squandered—to unlock a deeper understanding of infection-associated chronic illnesses and optimize preparedness for future pandemics. The world must rise to the occasion and address these challenges; the health and well-being of current and future generations depend on this. ■

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ACKNOWLEDGMENTS

Z.A.-A. carried out consulting (uncompensated) for Pfizer.

10.1126/science.adl0867

COMPUTER SCIENCE

Overcoming the noise in neural computing

Circuit strategies can enable noisy analog hardware to achieve high precision

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In the past 20 years, microelectronics technology has seen a tapering off of Moore's law (the exponential growth of transistor density on computer chips) and an end to Dennard scaling (wherein smaller transistors use correspondingly less power). To create computing systems that are more powerful, faster, and energy efficient for artificial intelligence (AI) and numerical computing applications, new approaches to computing are needed. Attaining these new capabilities not only requires exploration of new materials and devices, but as reported by Song *et al.* on page 903 of this issue (1), it also necessitates innovative circuits and new mathematical approaches.

Multicore computer architectures, which allow more than one process to operate at the same time, and a system-on-chip approach, which integrates all processes on a single computer chip, have supported massive applications such as large language models and other AI technologies, but at

great cost (2, 3). Scaling applications to larger sizes in conventional architectures requires information to move greater distances between memory and logic. Moreover, this movement has an energy cost. For example, for a single bit to move from one end of a chip to the other is on the order of a picojoule—10,000 times more than the energy to switch a transistor (4).

The separation of memory and logic in a traditional computer design (called the von Neumann architecture) enables program-mability, precision, and scaling to larger programs and datasets. New architectures must trade off something to gain efficiency because it is physically impossible to place enough memory near general-purpose computing to have the same benefits of a traditional design.

Two common approaches to overcome the von Neumann bottleneck—the slow movement of data between the memory and processor—make different trade-offs. Digital brain-like “spiking” architectures (neuromorphic computing) use low-precision but highly reliable event-driven communication, and analog crossbars use dense multibit memories but are limited by the noise inherent to

Challenges in computer architecture paradigms

Von Neumann designs separate processing from memory, creating a bottleneck in communication that is problematic with scaling. Analog crossbar architectures perform linear algebra operations, but as the number of crossbars increases, multiple noise sources affect computation precision. Spiking neuromorphic systems distribute computation over a large network but require new algorithms.

